



Thermogravimetric experiments based prediction of biomass pyrolysis behavior: A comparison of typical machine learning regression models in *Scikit-learn*

Yu Zhong^{a,b}, Fahang Liu^a, Guozhe Huang^a, Juan Zhang^a, Changhai Li^c, Yanming Ding^{a,b,*}

^a Faculty of Engineering, China University of Geosciences, Wuhan 430074, China

^b Institute for Natural Disaster Risk Prevention and Emergency Management, China University of Geosciences, Wuhan 430074, China

^c State Key Laboratory of Fire Science, University of Science and Technology of China, Hefei 230027, China

ARTICLE INFO

Keywords:

Biomass pyrolysis

Prediction

Machine learning

Comparison

Scikit-learn

ABSTRACT

A variety of machine learning (ML) models have been extensively utilized in predicting biomass pyrolysis owing to their prowess in deciphering complex non-linear relationships between inputs and outputs, but there is still a lack of consensus on the optimal methods. This study elaborates on the development, optimization, and evaluation of three ML methodologies, namely, artificial neural networks, random forest (RF), and support vector machines, aimed to determine the optimal model for accurate prediction of biomass pyrolysis behavior using thermogravimetric data. This work assesses the utility of thermal data derived from these models in the computation of kinetic and thermodynamic parameters, alongside an analysis of their statistical performance. Eventually, the RF model exhibits superior physical interpretability and the least discrepancy in predicting kinetic and thermodynamic parameters. Furthermore, a feature importance analysis conducted within the RF model framework quantitatively reveals that temperature and heating rate account for 98.5 % and 1.5 %, respectively.

1. Introduction

In the face of escalating global energy requirements and the diminishing reserves of fossil fuels, the shift towards eco-friendly and sustainable energy alternatives is gaining unprecedented importance (Ding et al., 2023). Biomass stands out as the most abundant and accessible source of renewable energy, offering potential for conversion into bio-fuels via a spectrum of thermochemical processes such as pyrolysis, combustion, and gasification (Ding et al., 2020b; Xu et al., 2020b). One key advantage of pyrolysis is its flexibility, allowing to produce a diverse array of chemicals and products depending on the process conditions (Bui et al., 2022; Khan et al., 2023). Compared with the other two techniques, pyrolysis distinguishes itself by yielding a diverse array of solid, liquid, and gaseous byproducts, which can subsequently be processed further through gasification or combustion (Ding et al., 2020a; Zhang et al., 2021; Zhang et al., 2022). Thus, investigating the pyrolysis behavior of biomass at high temperatures is crucial for utilizing biomass energy, optimizing process parameters, and designing more efficient equipment (Mankeed et al., 2022; Naveed et al., 2024).

Thermogravimetric analysis (TGA) stands as a cornerstone laboratory technique extensively employed in unraveling the intricate pyrolysis dynamics of diverse materials (Petrović et al., 2023; Zhang et al., 2022). By delineating the loss of mass relative to variations in temperature and time, TGA furnishes invaluable insights into thermal behavior (Wu et al., 2014). However, due to experimental limitations, predicting datasets spanning a wide range of heating rates poses a significant challenge. Consequently, researchers have pivoted towards computational methodologies, notably kinetic inverse modeling (Burra and Gupta, 2019; Richter and Rein, 2017) and machine learning (Ascher et al., 2022; Çepelioğullar et al., 2016, 2018; Kartal and Özveren, 2022; Zhong et al., 2024), to streamline laboratory procedures and predict thermal data. Kinetic inverse modeling typically necessitates a well-defined chemical kinetic reaction scheme (Zhong et al., 2023). However, achieving heightened prediction accuracy entails the adoption of intricate schemes teeming with multiple parameters, thus accentuating computational demands (Ding et al., 2019).

Therefore, the researchers wondered whether it would be possible to predict the pyrolysis behavior of biomass without knowing the reaction

* Corresponding author at: Faculty of Engineering, China University of Geosciences, Wuhan 430074, China.

E-mail address: dingym@cug.edu.cn (Y. Ding).

<https://doi.org/10.1016/j.marpolbul.2024.116361>

Received 23 January 2024; Received in revised form 7 April 2024; Accepted 7 April 2024

Available online 17 April 2024

0025-326X/© 2024 Elsevier Ltd. All rights reserved.

scheme (Muravyev et al., 2021). A lot of machine learning (ML) methods have been widely employed due to their ability to learn extremely non-linear input–output correlations (Chen et al., 2024; Luo et al., 2023). For example, artificial neural network (ANN) has demonstrated its value in solving various chemical engineering problems, but its application in pyrolysis, thermal analysis, and especially thermodynamic studies is still in its infancy (Ascher et al., 2022). Wang et al. (2024), Shafizadeh et al. (2024), and Balsora et al. (2023) employed ANN in their respective studies to forecast the high calorific value of biochar, the product distribution resulting from the co-pyrolysis of biomass and coal, and the pyrolysis products of biomass. On the other hand, because the decision tree is easy to follow and understand, the random forest (RF) based on the decision tree has also been applied in the field of biomass pyrolysis behavior prediction (Cheng et al., 2020; Yang et al., 2023). Zhu et al. (2019) used RF to predict biochar yield and carbon contents in biochar based on biomass properties and pyrolysis conditions. Liu et al. (2024) primarily utilized RF algorithms to forecast the pyrolysis conversion rate of various types of woody biomass, considering factors such as composition and heating rate. Additionally, some researchers have explored the use of support vector machine (SVM) as an alternative approach to modeling thermochemical processes (Cao et al., 2016; Chen et al., 2018; Elmaz et al., 2020; Shahbeig and Nosrati, 2020). Cao et al. (2016) and Chen et al. (2018) employed SVM to predict the distribution and characteristics of pyrolysis products, achieving very good statistical performance in their predictions. It is shown that ML can obtain prediction data with comparable accuracy or even better than kinetic inverse modeling without the need for specific pyrolysis reaction schemes and complex optimization algorithms (Çepeliogullar et al., 2016, 2018).

Despite the burgeoning utilization of ML methods in predicting biomass pyrolysis, there remains a lack of consensus regarding the optimal method. This study aims to explore the predictive performance of these three typical models (ANN, RF, SVM) that have been established and optimized based on the *scikit-learn* library in Python (Pedregosa et al., 2011a). Previous research (Çepeliogullar et al., 2018; Xing et al., 2019) predominantly relied on regression statistical performance metrics to evaluate machine learning models for predicting pyrolysis

behavior across training, validation, and test datasets. However, accurately capturing the physical significance inherent in pyrolysis data, particularly concerning thermogravimetric data, holds paramount importance. Hence, this paper adopts a methodological approach involving kinetic and thermodynamic analyses to compare the predictive efficacy of the three models. In essence, the outcomes of this research endeavor to furnish valuable insights for the selection and evaluation framework of machine learning models employed in forecasting biomass pyrolysis behavior.

2. Samples and methods

2.1. Samples

In the present study, beech wood (*Fagus sylvatica*) was selected as the sample for the thermogravimetric experiment. To eliminate free water, all beech samples were ground into powder and dried for approximately 24 h. The thermogravimetric experiment was carried out using a TA Instrument (SDT Q600 thermal analyzer) under a nitrogen atmosphere with eight heating rates (3, 5, 10, 20, 30, 40, 60, and 80 K/min). The experiment was conducted in the temperature range of 300 K to 1000 K, with the purge flow rate of pure nitrogen set at 100 mL/min. Fig. 1 shows the pyrolysis behaviors of beech wood at different heating rates in N₂.

The ML models were constructed using temperature and heating rate as inputs and mass loss as the output variable. For the ML modeling phase, a subset of 600 data points on mass loss, spanning temperatures from 300 K to 900 K at intervals of 1 K, was extracted from the experimental dataset recorded at each heating rate. Of these, 80 % of the data, corresponding to heating rates ranging from 3 to 60 K/min, were allocated for training the ML models, while the remaining 20 % were designated for validation purposes. Notably, data at an 80 K/min heating rate were exclusively utilized for testing to assess the ML models' generalization capabilities. The partitioning of data into training and validation sets was facilitated by the “*train_test_split*” function from the *scikit-learn* library (Pedregosa et al., 2011b) in Python.

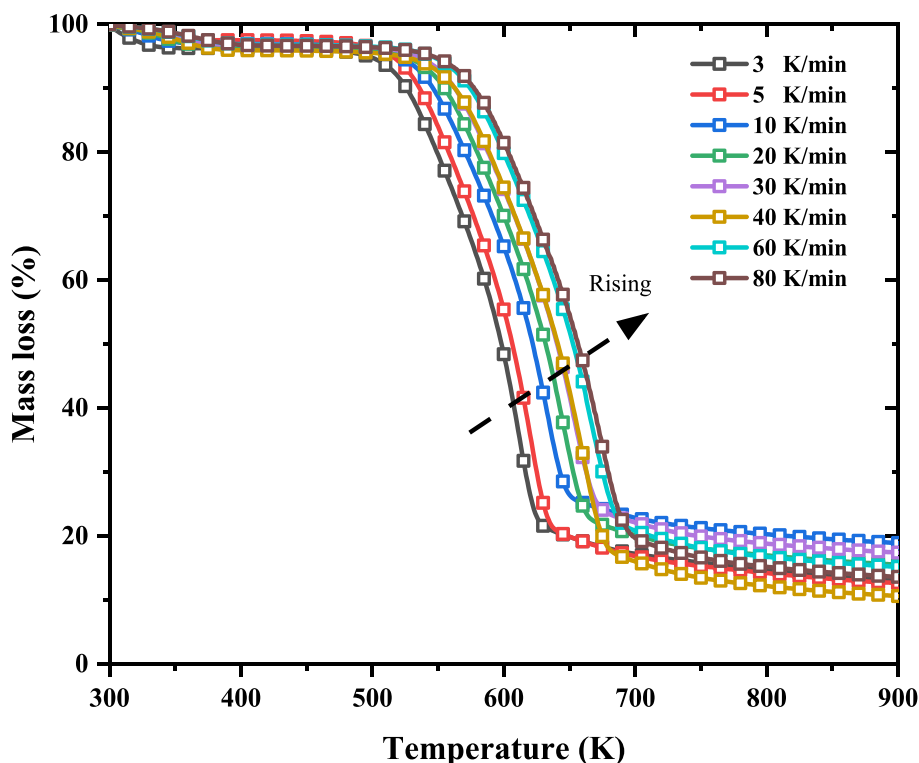


Fig. 1. Experimental mass loss with the temperature at different heating rates.

2.2. Machine learning regression models

2.2.1. Artificial neural network

The Artificial Neural Network (ANN), commonly referred to as a multi-layer perceptron, is characterized by its capacity to incorporate multiple hidden layers beyond the fundamental input and output layers. The architectural configuration of an ANN is depicted in Fig. 2, where x_i and o_i are the input and output vectors, f and g are the activation functions, w_i and b_i are the weight and bias vectors between different layers (Hecht-Nielsen, 1992). The weight and bias vectors are modified using the backpropagation algorithm until an error criterion or the number of iterations is reached.

2.2.2. Random forest

The Random Forest (RF) algorithm, an eminent ensemble learning technique, has garnered widespread acclaim within the data science and machine learning communities for its efficacy (Cutler et al., 2012). RF works by creating multiple decision trees for a given set of training samples, with each tree making its prediction. By combining the results of these individual trees using averaging, RF can enhance prediction accuracy and prevent overfitting, resulting in a more robust and reliable model. Fig. 3 delineates the schematic topology of the RF model, illustrating the integration of multiple decision trees to forge a consolidated output signal. This depiction underscores the synergy of the decision trees, culminating in a formidable composite model. In deploying RF, it is imperative to judiciously select hyperparameters, notably the forest's depth and the total count of trees, as these parameters significantly influence the model's performance (Xing et al., 2020).

2.2.3. Support vector machine

The support vector machine (SVM) method was developed by Vapnik (1999). The basic idea is to solve the separating hyperplane that can correctly divide the training data set and has the largest geometric interval. For the regression problem in this study, the optimization problem of support vector regression is as follows:

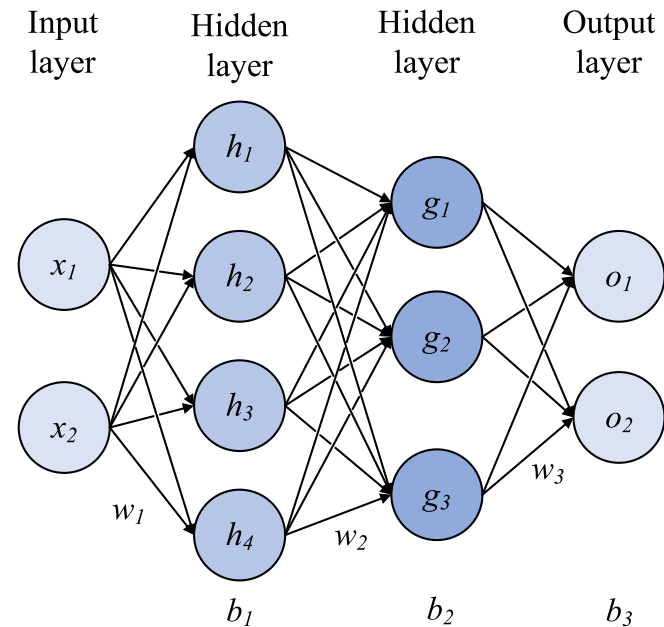


Fig. 2. The schematic topological architecture of ANN model.

$$\begin{aligned} \min_{\omega, b, \zeta} \quad & \frac{1}{2} \omega^T \omega + C \sum_{i=1}^l (\zeta_i + \zeta_i^*) \\ \text{s.t.} \quad & f_i - \omega^T \phi(\mathbf{x}_i) - \mathbf{b} \leq \varepsilon + \zeta_i, \\ & \omega^T \phi(\mathbf{x}_i) + \mathbf{b} - f_i \leq \varepsilon + \zeta_i^*, \\ & \zeta_i, \zeta_i^* \geq 0, i = 1, \dots, l \end{aligned} \quad (1)$$

where ω and $\mathbf{b}$ are the weight and bias vectors, respectively. $\Phi(\mathbf{x}_i)$ is the nonlinear function used to map the initial input vectors into a higher dimensional characteristic space. ζ_i and ζ_i^* are the regression loss. C is the penalty coefficient, which is the tolerance for error. The higher C is, the less error can be tolerated and the easier it is to overfit. The smaller C is, the easier it is to underfit. f_i is regression function, can be determined as Eq. (7) by introducing Lagrange multipliers:

$$f_i((\alpha_i - \alpha_i^*) \mathbf{K}(\mathbf{x}, z) + \mathbf{b}) \geq 1 - \zeta_i \quad (2)$$

where α_i and α_i^* are the Lagrange multipliers. $\mathbf{K}(\mathbf{x}, z)$ is the kernel function, and here this function is set as the common Gaussian radial basis function (RBF), which can be expressed as:

$$\mathbf{K}(\mathbf{x}, z) = \exp(-g \cdot d(\mathbf{x}, z)^2) \quad (3)$$

g is the width parameter. The above C and g , as the most typical hyperparameters of SVM, will be optimized later (Lei et al., 2019).

2.3. Optimization method

2.3.1. Data standardization

In ML, the diversity of input parameters frequently entails varying dimensions and units, exemplified by the heating rate and temperature in the current investigation. To neutralize the dimensional disparities among parameters, it is essential to undertake data standardization. This process ensures the parameters are rendered comparable, thereby facilitating the analysis and integration of multi-dimensional data (Zhu et al., 2019). Among them, the most typical is the normalization processing of data. The purpose of normalization is to limit the preprocessed data to a certain range, to eliminate the adverse effects caused by singular sample data. In this paper, the Z-score standardization method is used to standardize the input parameters, the formula is as follows:

$$x^* = \frac{x - \mu}{\sigma} \quad (4)$$

where μ is the mean of all sample data, and σ is the standard deviation of all sample data.

2.3.2. Grid search method based on k-fold cross validation

Grid search is a commonly used parameter adjustment method, which is an exhaustive method (Ma et al.; Syarif et al., 2016). The basic strategy is to give a series of hyperparameters, then iterate through all combinations of hyperparameters exhaustively, and select the optimal set of hyperparameters from all combinations. It is to find the optimal solution in all solutions by brute force method, which is also a process of training and comparison.

Cross validation involves the iterative utilization of data, partitioning the available sample dataset into various training and validation sets. This approach allows for the training of the model on one set of data and the subsequent evaluation of the model's predictive accuracy on another (Anguita et al.; Krstajic et al., 2014). This method differs significantly from the standard practice of dataset division, enabling the generation of multiple distinct training and validation sets. Consequently, a sample present in a training set during one iteration of the process may appear in a validation set in another, embodying the "cross" aspect of the technique. Specifically, k-fold cross-validation entails the computation of an average across k validations, facilitating the

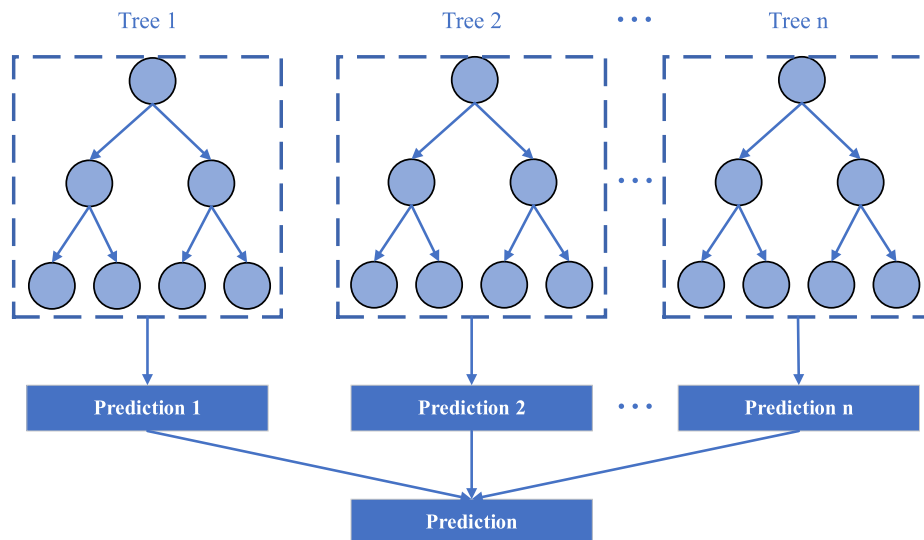


Fig. 3. The schematic topological architecture of RF model.

comparison of different combinations of hyperparameters as identified through grid search. It is noteworthy that hyper-parameter optimization of these models was performed using the *GridSearchCV* tool in Python's *scikit-learn* library (Pedregosa et al., 2011b), involving a series of trial-and-error tests.

2.4. Model performance

2.4.1. Statistical indicators

The Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Relative Root Mean Squared Error (RRMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2), which have been widely used to quantitatively compare the performances of various models in many previous ML studies, are defined here.

$$R^2 = 1 - \frac{\sum_{k=1}^n (y_{exp,k} - y_{pre,k})^2}{\sum_{k=1}^n (y_{exp,k} - \bar{y}_{exp,k})^2} \quad (5)$$

$$MSE = \frac{1}{n} \sum_{k=1}^n (y_{exp,k} - y_{pre,k})^2 \quad (6)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{k=1}^n (y_{exp,k} - y_{pre,k})^2} \quad (7)$$

$$MAE = \frac{1}{n} \sum_{k=1}^n |y_{exp,k} - y_{pre,k}| \quad (8)$$

$$RRMSE = \frac{RMSE}{\bar{y}_{exp}} \quad (9)$$

where, y_{pre} and y_{exp} are the predicted and experimental mass loss values, respectively, and n is the number of experimental data points for each experiment. $\bar{y}_{exp}$ is the mean of the experimental mass loss values.

2.4.2. Kinetics method

In order to study how applicable ML models predicted thermogravimetric data, a typical isoconversional kinetics method (Flynn–Wall–Ozawa method) (Flynn and Wall, 1966; Ozawa, 1965) was chosen to calculate the pyrolysis activation energies based on the experiment and prediction data. FWO method assumes the change in the apparent activation energy constant during the thermal process and

employs Doyle's (Doyle, 1961; Doyle, 1962) approximation to approach a solution for temperature integral. The reaction rate in logarithmic form can be expressed as:

$$\ln \beta = \ln \left(\frac{AE_a}{Rg(\alpha)} \right) - 5.331 - 1.052 \left(\frac{E_a}{RT} \right) \quad (10)$$

where β is the heating rate, R is the universal gas constant, and $g(\alpha)$ is the integral function of conversion. The slope (E_a/R) of the line determined from the relationship between $\ln \beta$ and $1/T$ can be used to calculate the activation energy E_a .

As one of the most popular model-fitting methods, Coats-Redfern (CR) method (Coats and Redfern, 1964) is obtained by asymptotic series expansion. It can be represented as:

$$\ln \frac{g(a)}{T^2} = \ln \left(\frac{AR}{\beta E_a} \right) - \frac{E_a}{RT} \quad (11)$$

By plotting $\ln(g(a)/T^2)$ versus $1/T$, activation energy E_a , and pre-exponential factor A can be ascertained from the slope and intercept of the fitting line, respectively. The pyrolysis mechanism and reaction model determine the equations of function $f(a)$ and $g(a)$. The $g(a)$ usually represents the reaction models, including nucleation and growth, phase-boundary controlled reaction, diffusion, and reaction nth-order model. The common reaction models of solid-state pyrolysis are shown in Table S1.

According to the recommendations of the International Confederation for Thermal Analysis and Calorimetry (Vyazovkin et al., 2011), the pre-exponential factor A should be calculated using the kinetic compensation effects (KCE), and can be described as:

$$\ln A_i = aE_i + b \quad (12)$$

where i is the different heating rates, a and b refer to constants for compensation effect, a is the intercept, and b is the slope of the regression line.

2.4.3. Thermodynamic method

The thermodynamic parameters include enthalpy change (ΔH), Gibb's free energy (ΔG), and entropy change (ΔS). The equations to determine these parameters are expressed below (Kim et al., 2010; Olszak-Humienik and Mozejko, 2000):

$$\Delta H = E_a - RT \quad (13)$$

$$\Delta G = E_a + RT_m \ln \left(\frac{K_B T_m}{hA} \right) \quad (14)$$

$$\Delta S = \frac{\Delta H - \Delta G}{T_m} \quad (15)$$

where T_m represents the average peak temperature on the mass loss curve, K_B is known as the Boltzmann constant (1.38×10^{-23} J/K). Then, h is referred to the Planck's constant and its value is 6.626×10^{-34} J·s.

3. Results and discussions

3.1. Comparison of statistical performance

3.1.1. Development and optimization of ANN model

The present section employed the default ANN regression model in the *scikit-learn* library of Python for training and prediction. Table 1 displays a selection of the default parameter configurations for the ANN. The statistical outcomes for the training, validation, and test datasets are depicted in Fig. 4. The results show that both the MSE and R^2 in the training set and validation set exceed 10^{-2} and are below 0.8, respectively. In particular, the R^2 of the test set is as low as 0.6257, indicating poor regression prediction performance of the default ANN model.

To refine the default artificial neural network (ANN) model's three critical hyperparameters—namely, the number of hidden layers, number of neurons, and solver—both data standardization and grid search optimization techniques were applied. Table 2 delineates the configurations for these parameters and their optimized values ascertained through 5-fold cross validation. Fig. 5 showcases the statistical achievements of the optimized ANN model. Compared with Fig. 4, a significant enhancement in performance is evident, particularly for the test set, where the mean squared error (MSE) is diminished from 10^{-2} to 10^{-3} , and R^2 escalates from 0.6257 to 0.9717. Detailed error analysis is shown in Table S5 of supplementary materials.

3.1.2. Development of RF model

As can be seen from the statistical performance of RF (Fig. 6), the predicted data agrees very well with the experimental data. Especially, in the test set, MSE is only 10^{-4} MSE and R^2 is 0.99, indicating the robust generalization capability of RF. It is worth noting that in the default RF model, the dataset is not normalized and the hyperparameters are not optimized. The depth of the forest grows freely with training until good performance is obtained, and the number of trees defaults to 100, which are the two most important hyperparameters of RF.

The superior non-linear regression performance of RF is attributed to its ensemble approach, which integrates multiple decision trees, employs randomization in the selection of samples and features, and leverages the intrinsic non-linear processing capabilities of individual decision trees (Cutler et al., 2012). Within an RF model, each decision tree functions as a classifier and the aggregation of their outcomes mitigates the constraints associated with conventional linear regression models, thereby augmenting the model's capacity for non-linear fitting and enhancing predictive accuracy. Additionally, RF enhances model robustness against overfitting by randomly choosing a subset of the training dataset for the formation of each decision tree's training set. This methodology underpins the remarkable predictive efficacy observed in even the default, non-optimized RF models. Detailed error

analysis is shown in Table S6 of supplementary materials.

3.1.3. Development and optimization of SVM model

The statistical performance of the support vector machine default model is shown in Fig. 7. C and g , the two key hyperparameters of SVM model, are set to 1 and “scale” by default, respectively. The expression of $scale$ is as follows:

$$\gamma = \frac{1}{m \cdot X} \quad (16)$$

where m is the number of sample features and X is the variance of the sample features. It can be found that although the statistical performance (MSE and R^2) of the default SVM model is excellent, its regression points are not concentrated, mainly because the error values are averaged.

The optimization strategy applied to the SVM model mirrors that utilized for the ANN model, with a particular focus on fine-tuning the two critical hyperparameters: C and g . These parameters were adjusted across ranges from 1 to 20 for C , and from 10^{-5} to 10^3 for g , employing step increments of 1 and 10^1 respectively. Fig. 8 illustrates the regression statistics performance of the optimized SVM, revealing improved performance on both the training and validation sets. However, the performance did not improve when tested on an independent set, and there was a significant occurrence of “overfitting”. The results demonstrate that optimizing SVM on the dataset employed in this study presented a challenge. Consequently, exploring alternative, more powerful optimization methods is vital for SVM. Detailed error analysis is shown in Table S7 of supplementary materials.

3.1.4. Comparison of statistical performance with previous studies

In this section, the statistical errors corresponding to the ML model in this paper are compared with those in previous studies. A comprehensive comparison has been conducted with regard to the principal statistical indicators of accuracy, specifically: MSE, RMSE, RRMSE, MAE, and R^2 . The detailed comparative analysis is encapsulated in Table 3, illustrating the performance metrics for the evaluation of predictive models under consideration.

The results show that the RF model constructed in this study is close to the R^2 of the models established by Wei et al. (2023), Dubdub and Al-Yaari (2020), Çepeliogullar et al. (2016) and Shahbeig and Nosrati (2020). In terms of MSE, the performance of this paper is better than Wei et al. (2023), Dubdub and Al-Yaari (2020), Çepeliogullar et al. (2016), and slightly lower than Shahbeig and Nosrati (2020). An important purpose of this paper is to compare the predictive performance of different machine learning. According to the discussions in Section 3.1.1–3.1.3, it can be found that RF model has the best performance, and the results of Wei et al. also support this conclusion to a certain extent. In general, the performance of the model established in this study is basically the same as that of previous studies, or even better.

3.2. Prediction performance based on kinetic analysis

Beyond evaluating the statistical regression performance of the models, the study also delved into the physical significance inherent in the predicted data. Fig. 9 presents a comparative analysis of the experimental thermogravimetric data alongside the predictions produced by the three ML models developed and optimized in this study. Despite the regression coefficients of the models surpassing the 0.9 threshold, discernible disparities in their predictive capabilities are evident. Remarkably, the ANN and RF models exhibit notably superior performance. This discrepancy not only highlights their strengths but also underscores the limitations associated with solely focusing on the statistical metrics of model predictions, advocating for a more holistic approach that encompasses both statistical and physical considerations.

Furthermore, the prediction data of ANN and RF with better

Table 1
The settings of the partial parameters for default ANN.

Parameter	The default value
hidden_layer_sizes	(100,)
activation	relu
solver	adam
max_iter	200

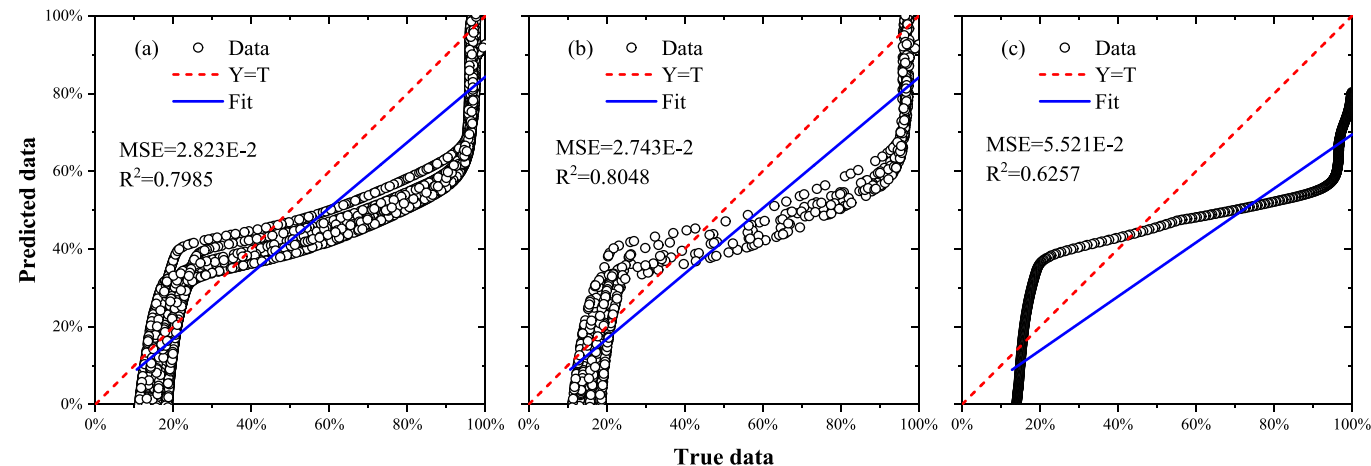


Fig. 4. Regression statistical performances of training (a), validation (b), and test (c) data-sets for default ANN.

Table 2
The range and optimized value of grid search for ANN model.

Parameter	Range	Optimized value
Hidden layers	1, 2, 3, 4	(20,20,20,20)
Neurons	5, 10, 15, 20	
Solver	adam, sgd, lbfgs	adam

performance were analyzed, and the conversion rate data was obtained. The prediction data of the two models were consistent with the experimental values as can be seen from Fig. 10. Then, kinetic analysis was performed by isoconversion rate method (FWO), which is described in Section 2.4.2. The linear fitting equations derived from the FWO method are illustrated in Fig. 11, with the calculated activation energies and their corresponding R² values presented in Table 4. Compared with the activation energy calculated from experimental data, it is observed that although the average activation energy of the two models is not

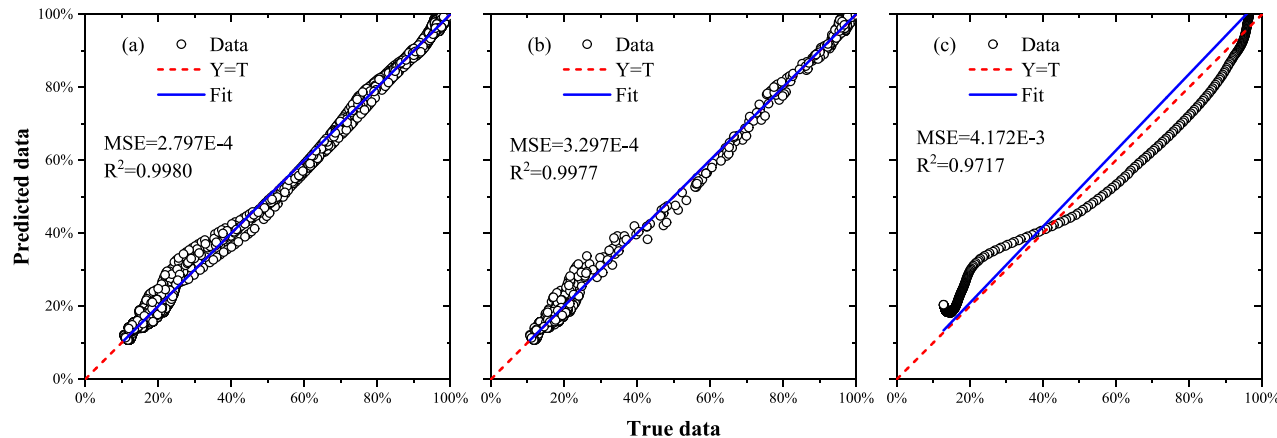


Fig. 5. Regression statistical performances of training (a), validation (b), and test (c) data-sets for optimized ANN.

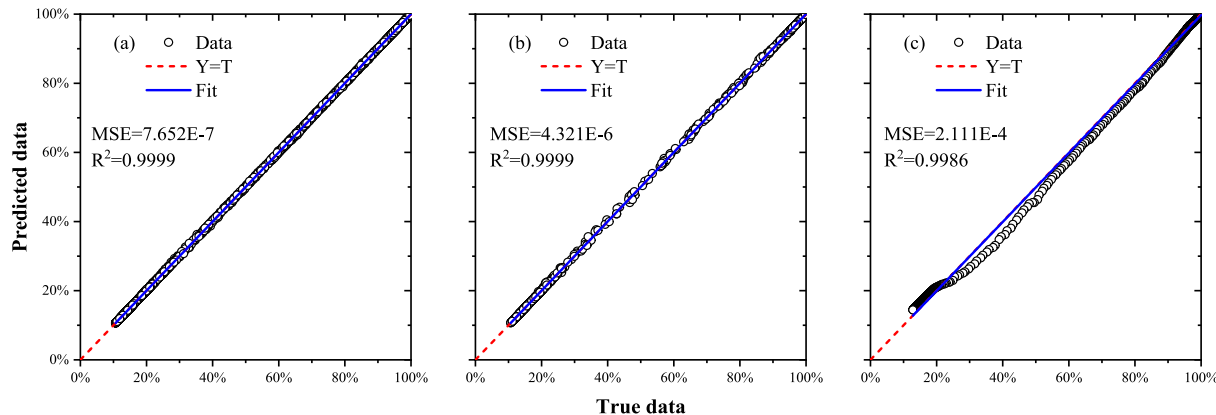


Fig. 6. Regression statistical performances of training (a), validation (b), and test (c) data-sets for default RF.

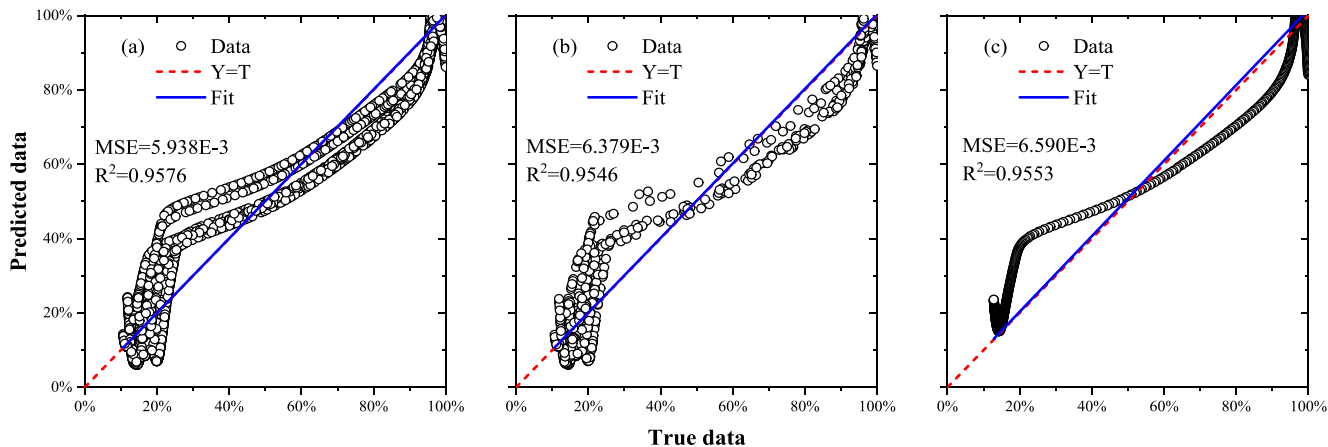


Fig. 7. Regression statistical performances of training (a), validation (b), and test (c) data-sets for default SVM.

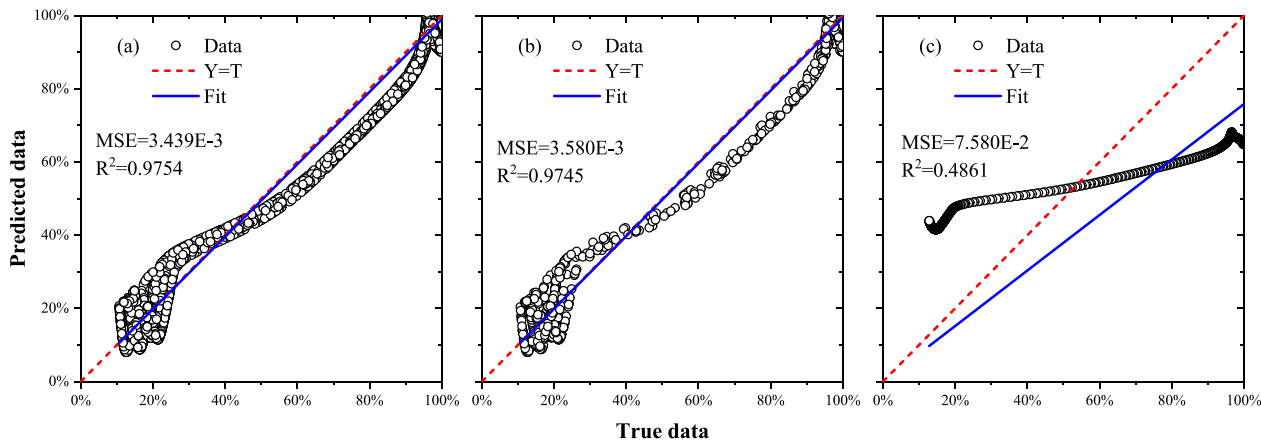


Fig. 8. Regression statistical performances of training (a), validation (b), and test (c) data-sets for optimized SVM.

Table 3
Comparison of statistical performance with previous studies.

Model	Material	MSE	RMSE	RRMSE	MAE	R ²	Reference
default RF	beech	2.11E-4	1.45E-2	1.11E-2	2.65E-2	0.9986	This study
Extra-Trees	tobacco	–	1.74E-2	–	–	0.9730	Wei et al. (2023)
ANN	polyethylene	–	1.70E-1	–	7.94E-2	–	Dubdub and Al-Yaari (2020)
ANN	refuse-derived fuel	3.74E-2	–	–	–	0.9998	Çepeliogullar et al. (2016)
SVM	bacterial biomass	5.25E-5	–	–	–	0.9999	Shahbeig and Nosrati (2020)

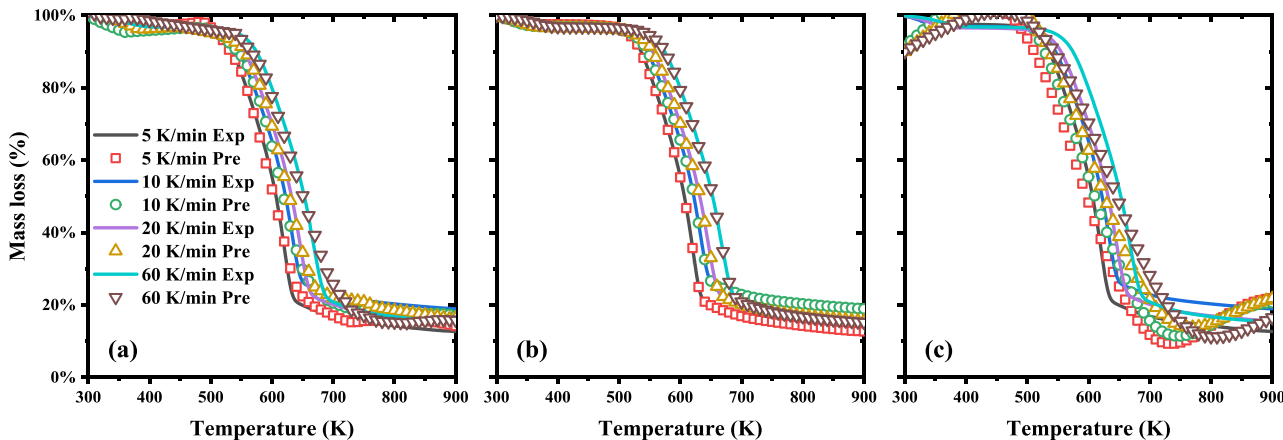


Fig. 9. Comparison of TG data predicted by optimized ANN (a), default RF (b), and default SVM (c) with experimental data at different heating rates.

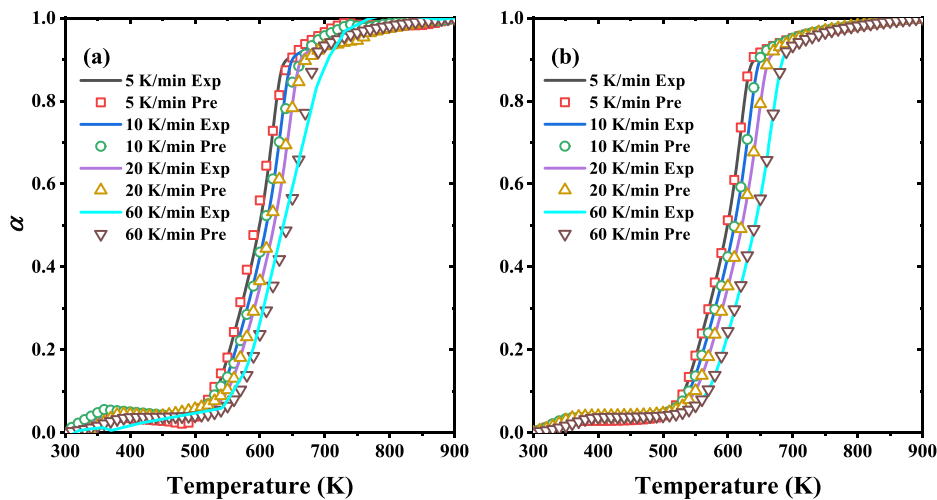


Fig. 10. Comparison of conversion rates based on predicted data by ANN (a) and RF (b) with experimental data at different heating rates.

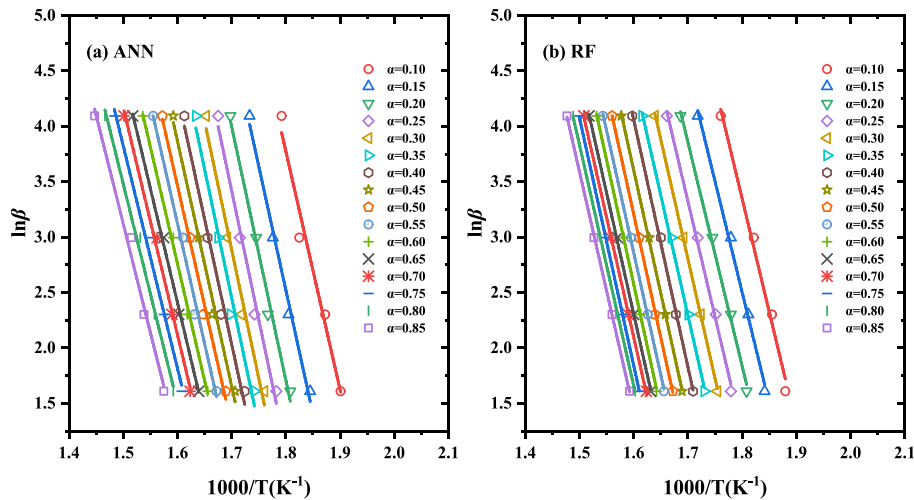


Fig. 11. FWO plots for different conversion rates based on predicted TG data by ANN (a) and RF (b).

Table 4
Comparison of activation energy (kJ/mol) calculated by FWO method for predicting TG data between RF and ANN.

α	Exp		ANN			RF		
	E_a	R^2	E_a	R^2	%Error	E_a	R^2	%Error
0.10	161.43	0.9910	169.90	0.9595	−4.98	161.43	0.9865	0.00
0.15	157.78	0.9982	176.68	0.9869	−10.70	158.18	0.9942	−0.25
0.20	159.50	0.9970	180.97	0.9826	−11.87	159.50	0.9955	0.00
0.25	164.54	0.9970	183.92	0.9803	−10.53	164.54	0.9955	0.00
0.30	170.22	0.9995	183.40	0.9727	−7.19	170.22	0.9993	0.00
0.35	170.80	0.9988	183.54	0.9679	−6.94	170.80	0.9982	0.00
0.40	175.39	0.9988	178.49	0.9764	−1.74	176.28	0.9982	−0.50
0.45	178.80	0.9994	174.72	0.9841	2.33	175.77	0.9995	1.73
0.50	179.80	0.9996	171.00	0.9883	5.15	176.01	0.9994	2.15
0.55	174.26	0.9995	171.10	0.9858	1.85	174.26	0.9993	0.00
0.60	176.36	0.9996	166.49	0.9947	5.93	177.06	0.9992	−0.39
0.65	175.48	0.9995	162.80	0.9965	7.79	172.74	0.9994	1.59
0.70	174.30	1.0000	161.93	0.9933	7.64	170.89	0.9995	1.99
0.75	173.01	0.9999	158.65	0.9890	9.05	176.48	0.9999	−1.97
0.80	175.16	0.9999	156.54	0.9890	11.90	171.87	0.9999	1.92
0.85	177.88	0.9999	154.90	0.9824	14.83	170.08	0.9997	4.58
Mean	171.54	—	170.94	0.9831	—	170.38	0.9977	—

significantly different from the experimental value, there is an absolute error of more than 10 % for ANN at a specific conversion rate, while RF has an error of less than 5 % at each conversion rate. Despite both

models demonstrating excellent statistical regression performance, RF outperforms in predicting the physical interpretability of the data. Kinetic compensation effects (KCE) have been widely reported for

many different thermochemical processes, including combustion, pyrolysis, and gasification. The kinetics parameters of three data sets at heating rate of 5, 10, 20, and 60 K/min by CR method are listed in Tables S2, S3 and S4. Based on the kinetic parameters obtained by CR method, compensation effect constants can be calculated at various heating rates, which can be used to estimate the pre-exponential factor (A) of three data sets (experimental data, ANN, and RF predicted TG data). Fig. 12 demonstrates the KCE parameters linearity correlations of three data sets at the heating rates of 5, 10, 20 and 60 K/min. Ultimately, the activation energy calculated based on the FWO method was incorporated into the KCE equation for each dataset, and the corresponding pre-exponential factors were obtained (Table 5). The kinetic parameters calculated by the ML models established in this paper are not only consistent with the experimental data but also little different from the results of the existing pyrolysis literature of woody biomass materials (Table 6).

3.3. Prediction performance based on thermodynamic analysis

The assessment of the predictive proficiency of the ML models under discussion extends to thermodynamic parameters, in addition to kinetic parameters. Kinetic parameters describe the rates and mechanisms of pyrolysis reactions, whereas thermodynamic parameters provide information regarding energy changes and their feasibility. Integrating both parameters allow for a holistic comprehension of the reaction rate and process in biomass pyrolysis, thereby aiding in the optimization of the design and control of the biomass pyrolysis process. Table 6 presents the comparison results of the thermodynamic parameters calculated from the experimental data and the ANN and RF prediction data. It can be found that the performance of the two models is not much different in terms of thermodynamic parameters, but for ΔH , RF is closer to the experimental value than ANN. The reason is that according to its calculation expression, in addition to the excellent performance of the kinetic parameter activation energy, it is also related to the corresponding temperature at different conversion rates.

Moreover, compared with the existing literature (Table 8), the ML models established in this paper not only agree well with the experimental data, but also have little difference with the thermodynamic parameters of other woody biomass materials (Tables 7 and 8).

3.4. Feature importance

Moreover, this study represents an effort to ascertain the significance of two input features, specifically the heating rate and temperature, in forecasting thermogravimetric mass loss data. The comparison of mathematical and physical performance underscores the RF model's superior predictive capabilities. A feature importance analysis conducted within the RF framework quantitatively reveals that temperature accounts for 98.5 % of the predictive accuracy, while the heating rate

Table 5

Comparison of pre-exponential factor ($\ln A$ (min^{-1})) calculated by KCE from the experimental data, ANN, and RF predicted TG data.

α	Exp	ANN	RF
0.10	17.36	20.44	18.48
0.15	16.65	21.75	17.85
0.2	16.99	22.58	18.11
0.25	17.97	23.15	19.09
0.3	19.07	23.05	20.19
0.35	19.19	23.07	20.30
0.40	20.08	22.10	21.37
0.45	20.74	21.37	21.27
0.50	20.94	20.65	21.31
0.55	19.86	20.67	20.97
0.60	20.27	19.78	21.52
0.65	20.10	19.06	20.68
0.70	19.87	18.89	20.32
0.75	19.62	18.26	21.41
0.80	20.04	17.85	20.51
0.85	20.56	17.53	20.16
Mean	19.33	20.64	20.22

contributes to 1.5 %. This discrepancy underscores the predominant influence of temperature over the heating rate on mass loss, a finding corroborated by TG curve observations. Such an analysis furnishes a quantitative insight into the parameters that significantly impact mass loss in TGA experiments, enhancing the understanding and prediction of pyrolysis behavior.

The Technology Readiness Level (TRL) is a system used to estimate the maturity of a technology. The scale ranges from 1 (basic principles observed) to 9 (actual system proven in operational environment), as follows: basic principles (TRL 1–2), theoretical development (TRL 3), component validation in a laboratory setting (TRL 4), system/subsystem model or prototype demonstration in a related environment (TRL 5 or 6), system prototype demonstration in an operational environment (TRL 7), actual system proven in operational environment (TRL 8–9) (Beims et al., 2019). This study compares the performance of different ML models in predicting biomass pyrolysis at the laboratory scale and identifies the best model, providing kinetics and thermodynamics information for industrial-scale pyrolysis, belonging to TRL 3–4.

4. Conclusions

This study presents a detailed comparative analysis of three machine learning models for predicting the pyrolysis behavior of biomass, utilizing thermogravimetric data. Beyond merely comparing the default settings of each model, advanced optimization techniques such as grid search algorithms and k-fold cross-validation were employed to enhance the performance of models that initially exhibited suboptimal results. Notably, the statistical regression performance, quantified by the coefficient of determination (R^2), for all three models on the test dataset

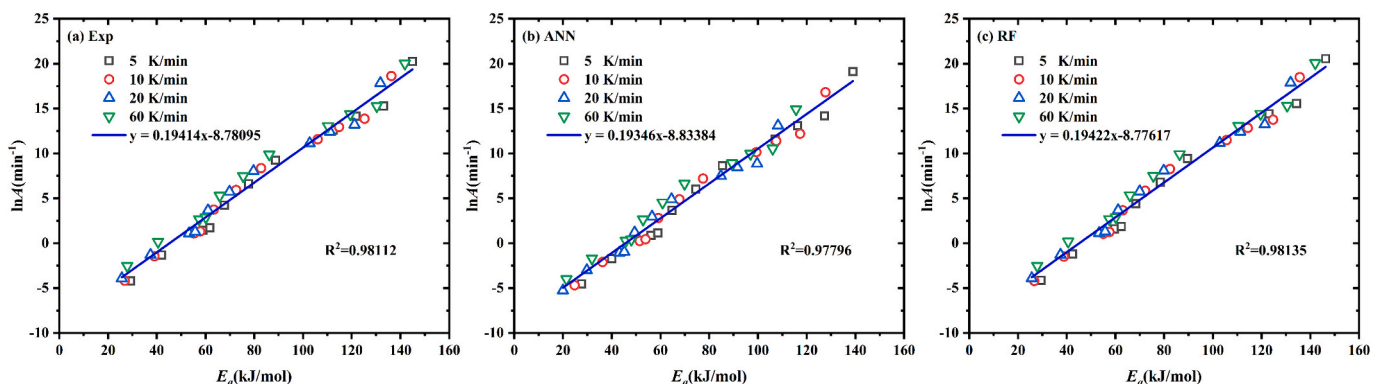


Fig. 12. The KCE correlations ($\ln A$ vs. E_a) of different data obtained by Coats-Redfern method: (a) experimental data, (b) RF, (c) ANN.

Table 6
Comparison of kinetics results with existing literature.

	Exp	ANN	RF	Tabal et al. (2021)	Di Blasi and Branca (2001)	Branca et al. (2005)
<i>Ea</i> (kJ/mol)	171.54	170.94	170.38	172.31 ± 3	141.2 ± 15.8	148.6
ln <i>A</i> (s ^{−1})	19.33	20.64	20.22	31.84	22.2 ± 2.9	23.40

Table 7
Comparison of thermodynamic parameters calculated from the experimental data, ANN, and RF predicted TG data.

a	ΔH (kJ/mol)			ΔG (kJ/mol)			ΔS (kJ/mol-K)		
	Exp	ANN	RF	Exp	ANN	RF	Exp	ANN	RF
0.10	156.88	165.40	156.88	230.21	219.73	223.80	−0.11	−0.09	−0.10
0.15	153.12	172.03	153.53	230.35	219.80	223.90	−0.12	−0.08	−0.11
0.20	154.75	176.23	154.75	230.28	219.85	223.86	−0.12	−0.07	−0.11
0.25	159.73	179.11	159.73	230.08	219.88	223.70	−0.11	−0.07	−0.10
0.30	165.33	178.53	165.33	229.85	219.88	223.52	−0.10	−0.07	−0.09
0.35	165.84	178.61	165.84	229.82	219.88	223.49	−0.10	−0.07	−0.09
0.40	170.37	173.50	171.27	229.64	219.82	223.32	−0.09	−0.08	−0.08
0.45	173.72	169.68	170.69	229.50	219.79	223.33	−0.09	−0.08	−0.08
0.50	174.67	165.90	170.88	229.46	219.74	223.33	−0.09	−0.09	−0.08
0.55	169.08	165.95	169.08	229.69	219.74	223.39	−0.09	−0.09	−0.08
0.60	171.13	161.29	171.83	229.60	219.70	223.30	−0.09	−0.09	−0.08
0.65	170.21	157.55	167.47	229.64	219.66	223.43	−0.09	−0.10	−0.09
0.70	169.00	156.62	165.59	229.68	219.64	223.50	−0.09	−0.10	−0.09
0.75	167.67	153.29	171.15	229.73	219.61	223.32	−0.10	−0.11	−0.08
0.80	169.79	151.12	166.50	229.65	219.59	223.46	−0.09	−0.11	−0.09
0.85	172.46	149.42	164.67	229.53	219.57	223.52	−0.09	−0.11	−0.09
Mean	166.49	165.89	165.32	229.79	219.74	223.51	−0.10	−0.09	−0.09

Table 8
Comparison of thermodynamics results with existing literature.

	Exp	ANN	RF	Xu et al. (2020a)	Tabal et al. (2021)
ΔH (kJ/mol)	166.49	165.89	165.32	176.11	170.14
ΔG (kJ/mol)	229.79	219.74	223.51	198.39	172.43
ΔS (kJ/mol-K)	−0.10	−0.09	−0.09	−34.21	−4.08

exceeded the threshold of 0.9, underscoring their robust predictive capabilities. Among the evaluated models, the Random Forest (RF) algorithm, even with its default parameters, demonstrated superior physical interpretability and the lowest deviation in predicting kinetic and thermodynamic parameters. Furthermore, a meticulous feature importance analysis within the RF model framework revealed that temperature and heating rate were responsible for 98.5 % and 1.5 % of the model's predictive power, respectively, highlighting their critical roles in biomass pyrolysis prediction.

CRedit authorship contribution statement

Yu Zhong: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Fahang Liu:** Formal analysis, Validation. **Guozhe Huang:** Formal analysis, Validation. **Juan Zhang:** Writing – review & editing, Writing – original draft, Data curation. **Changhai Li:** Writing – review & editing, Writing – original draft. **Yanming Ding:** Writing – review & editing, Writing – original draft, Validation, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors would like to acknowledge financial support sponsored by National Natural Science Foundation of China (No. 52376132), Natural Science Foundation of Hubei Province of China, and Science and Technology Research Program of the Department of Education in Hubei Province, China (No. B2021001).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2024.116361>.

References

Anguita, D., Ghio, A., Ridella, S., Sterpi, D., K-Fold Cross Validation for Error Rate Estimate in Support Vector Machines. pp. 291–297.

Ascher, S., Watson, I., You, S., 2022. Machine learning methods for modelling the gasification and pyrolysis of biomass and waste. *Renew. Sust. Energ. Rev.* 155, 111902.

Balsora, H.K., S. K., Joshi, J.B., Sharma, A., Chakinala, A.G., 2023. Artificial neural network-based models for the prediction of biomass pyrolysis products from preliminary analysis. *Ind. Eng. Chem. Res.* 62 (36), 14311–14319.

Beims, R.F., Simonato, C.L., Wiggers, V.R., 2019. Technology readiness level assessment of pyrolysis of trygliceride biomass to fuels and chemicals. *Renew. Sust. Energ. Rev.* 112, 521–529.

Branca, C., Albano, A., Di Blasi, C., 2005. Critical evaluation of global mechanisms of wood devolatilization. *Thermochim. Acta* 429 (2), 133–141.

Bui, H.H., Tran, K.Q., Luengnaruemitchai, A., Seisenbaeva, G., Naqvi, S.R., Chen, W.H., 2022. Valorisation of Napier grass and agricultural residues via pyrolysis- a kinetic study. *Chem. Eng. Trans.* 92, 169–174.

Burra, K.R.G., Gupta, A.K., 2019. Modeling of biomass pyrolysis kinetics using sequential multi-step reaction model. *Fuel* 237, 1057–1067.

Cao, H., Xin, Y., Yuan, Q., 2016. Prediction of biochar yield from cattle manure pyrolysis via least squares support vector machine intelligent approach. *Bioresour. Technol.* 202, 158–164.

Çepeliogullar, Ö., Mutlu, İ., Yaman, S., Haykiri-Acma, H., 2016. A study to predict pyrolytic behaviors of refuse-derived fuel (RDF): artificial neural network application. *J. Anal. Appl. Pyrolysis* 122, 84–94.

- Çepeliogullar, Ö., Mutlu, İ., Yaman, S., Haykiri-Acma, H., 2018. Activation energy prediction of biomass wastes based on different neural network topologies. *Fuel* 220, 535–545.
- Chen, X., Zhang, H., Song, Y., Xiao, R., 2018. Prediction of product distribution and bio-oil heating value of biomass fast pyrolysis. *Chem. Eng. Process. Process Intensif.* 130, 36–42.
- Chen, F., Ding, L., Zhu, Y., Ren, G., Man, Y., Hong, K., Lang, L., StrC6m, H., Xiong, Q., 2024. Comprehensive kinetic modeling and product distribution for pyrolysis of pulp and paper mill sludge. *Sci. Total Environ.* 924, 171665.
- Cheng, F., Luo, H., Colosi, L.M., 2020. Slow pyrolysis as a platform for negative emissions technology: an integration of machine learning models, life cycle assessment, and economic analysis. *Energy Convers. Manag.* 223, 113258.
- Coats, A.W., Redfern, J.P., 1964. Kinetic parameters from thermogravimetric data. *Nature* 201 (4914), 68–69.
- Cutler, A., Cutler, D.R., Stevens, J.R., 2012. Random forests. In: Zhang, C., Ma, Y. (Eds.), *Ensemble Machine Learning: Methods and Applications*. Springer, New York, New York, NY, pp. 157–175.
- Di Blasi, C., Branca, C., 2001. Kinetics of primary product formation from wood pyrolysis. *Ind. Eng. Chem. Res.* 40 (23), 5547–5556.
- Ding, Y., Zhang, J., He, Q., Huang, B., Mao, S., 2019. The application and validity of various reaction kinetic models on woody biomass pyrolysis. *Energy* 179, 784–791.
- Ding, Y., Fukumoto, K., Ezekoye, O.A., Lu, S., Wang, C., Li, C., 2020a. Experimental and numerical simulation of multi-component combustion of typical charring material. *Combust. Flame* 211, 417–429.
- Ding, Y., Huang, B., Li, K., Du, W., Lu, K., Zhang, Y., 2020b. Thermal interaction analysis of isolated hemicellulose and cellulose by kinetic parameters during biomass pyrolysis. *Energy* 195, 117010.
- Ding, Y., Jiang, G., Fukumoto, K., Zhao, M., Zhang, X., Wang, C., Li, C., 2023. Experimental and numerical simulation of multi-component combustion of typical no-charring material. *Energy* 262, 125555.
- Doyle, C.D., 1961. Kinetic analysis of thermogravimetric data. *J. Appl. Polym. Sci.* 5 (15), 285–292.
- Doyle, C.D., 1962. Estimating isothermal life from thermogravimetric data. *J. Appl. Polym. Sci.* 6 (24), 639–642.
- Dubdub, I., Al-Yaari, M., 2020. Pyrolysis of Low Density Polyethylene: Kinetic Study Using TGA Data and ANN Prediction. *Polymers*.
- Elmaz, F., Yücel, Ö., Mutlu, A.Y., 2020. Predictive modeling of biomass gasification with machine learning-based regression methods. *Energy* 191, 116541.
- Flynn, J.H., Wall, L.A., 1966. A quick, direct method for the determination of activation energy from thermogravimetric data. *J. Polym. Sci. B Polym. Lett.* 4 (5), 323–328.
- Hecht-Nielsen, R., 1992. III.3 - theory of the backpropagation neural network. In: Wechsler, H. (Ed.), *Neural Networks for Perception*. Academic Press, pp. 65–93.
- Kartal, F., Özveren, U., 2022. Prediction of activation energy for combustion and pyrolysis by means of machine learning. *Therm. Sci. Eng. Prog.* 33, 101346.
- Khan, A., Ali, I., Naqvi, S.R., AlMhamadi, H., Shahbaz, M., Ali, A.M., Shahzad, K., 2023. Assessment of thermokinetic behaviour of tannery sludge in slow pyrolysis process through artificial neural network. *Chemosphere* 337, 139226.
- Kim, Y.S., Kim, Y.S., Kim, S.H., 2010. Investigation of thermodynamic parameters in the thermal decomposition of plastic waste-waste lube oil compounds. *Environ. Sci. Technol.* 44 (13), 5313–5317.
- Krstajic, D., Buturovic, L.J., Leahy, D.E., Thomas, S., 2014. Cross-validation pitfalls when selecting and assessing regression and classification models. *J. Chemother.* 6 (1), 10.
- Lei, C., Deng, J., Cao, K., Xiao, Y., Ma, L., Wang, W., Ma, T., Shu, C., 2019. A comparison of random forest and support vector machine approaches to predict coal spontaneous combustion in gob. *Fuel* 239, 297–311.
- Liu, J., Lyu, H., Cheng, C., Xu, Z., Zhang, W., 2024. Enhancing pyrolysis process monitoring and prediction for biomass: a machine learning approach. *Fuel* 362, 130873.
- Luo, H., Wang, X., Liu, X., Yi, L., Wu, X., Yu, X., Ouyang, Y., Liu, W., Xiong, Q., 2023. Machine learning based prediction of biomass pyrolysis with detailed reaction kinetics for thermally-thick particles: from 1D to 0D. *Chem. Eng. Sci.* 280, 119060.
- Ma, X., Zhang, Y., Wang, Y., Performance evaluation of kernel functions based on grid search for support vector regression. *IEEE*, pp. 283–288.
- Mankeed, P., Onsree, T., Naqvi, S.R., Shimpalee, S., Tippayawong, N., 2022. Kinetic and thermodynamic analyses for pyrolysis of hemp hurds using discrete distributed activation energy model. *Case Stud. Therm. Eng.* 31, 101870.
- Muravyev, N.V., Luciano, G., Ornaghi, H.L., Svoboda, R., Vyazovkin, S., 2021. Artificial neural networks for pyrolysis, thermal analysis, and thermokinetic studies: the status quo. *Molecules* 26 (12), 3727.
- Naveed, M.H., Khan, M.N.A., Mukarram, M., Naqvi, S.R., Abdullah, A., Haq, Z.U., Ullah, H., Mohamadi, H.A., 2024. Cellulosic biomass fermentation for biofuel production: review of artificial intelligence approaches. *Renew. Sust. Energ. Rev.* 189, 113906.
- Olszak-Humienik, M., Mozejko, J., 2000. Thermodynamic functions of activated complexes created in thermal decomposition processes of sulphates. *Thermochim. Acta* 344 (1), 73–79.
- Ozawa, T., 1965. A new method of analyzing thermogravimetric data. *Bull. Chem. Soc. Jpn.* 38 (11), 1881–1886.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., 2011a. Scikit-learn: machine learning in python. *J. Mach. Learn. Res.* 12, 2825–2830.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, E., 2011b. Scikit-learn: machine learning in Python. *J. Mach. Learn. Res.* 12, 2825–2830.
- Petrović, A., Stergar, J., Skodić, L., Rašl, N., Cencić Predikaka, T., Čuček, L., Goricanec, D., Urbanc, D., 2023. Thermo-kinetic analysis of pyrolysis of thermally pre-treated sewage sludge from the food industry. *Therm. Sci. Eng. Prog.* 42, 101863.
- Richter, F., Rein, G., 2017. Pyrolysis kinetics and multi-objective inverse modelling of cellulose at the microscale. *Fire Saf. J.* 91, 191–199.
- Shafizadeh, A., Shahbeik, H., Rafiee, S., Fardi, Z., Karimi, K., Peng, W., Chen, X., Tabatabaei, M., Aghbashlo, M., 2024. Machine learning-enabled analysis of product distribution and composition in biomass-coal co-pyrolysis. *Fuel* 355, 129464.
- Shahbeig, H., Nosrati, M., 2020. Pyrolysis of biological wastes for bioenergy production: Thermo-kinetic studies with machine-learning method and Py-GC/MS analysis. *Fuel* 269, 117238.
- Syarif, I., Prugel-Bennett, A., Wills, G., 2016. SVM parameter optimization using grid search and genetic algorithm to improve classification performance. *TELKOMNIKA (Telecommun. Comput. Electron. Control)* 14 (4), 1502–1509.
- Tabal, A., Barakat, A., Aboulkas, A., El harfi, K., 2021. Pyrolysis of ficus nitida wood: determination of kinetic and thermodynamic parameters. *Fuel* 283, 119253.
- Vapnik, V.N., 1999. An overview of statistical learning theory. *IEEE Trans. Neural Netw.* 10 (5), 988–999.
- Vyazovkin, S., Burnham, A.K., Criado, J.M., Perez-Maqueda, L.A., Popescu, C., Sbirrazzuoli, N., 2011. ICTAC Kinetics Committee recommendations for performing kinetic computations on thermal analysis data. *Thermochim. Acta* 520 (1), 1–19.
- Wang, M., Xie, Y., Gao, Y., Huang, X., Chen, W., 2024. Machine learning prediction of higher heating value of biochar based on biomass characteristics and pyrolysis conditions. *Bioresour. Technol.* 395, 130364.
- Wei, H., Xing, J., Luo, K., Peng, Y., Fan, J., Zhang, K., Wang, H., 2023. Predicting tobacco pyrolysis based on chemical constituents and heating conditions using machine learning approaches. *Fuel* 335, 126895.
- Wu, K., Liu, J., Wu, Y., Chen, Y., Li, Q., Xiao, X., Yang, M., 2014. Pyrolysis characteristics and kinetics of aquatic biomass using thermogravimetric analyzer. *Bioresour. Technol.* 163, 18–25.
- Xing, J., Luo, K., Wang, H., Gao, Z., Fan, J., 2019. A comprehensive study on estimating higher heating value of biomass from proximate and ultimate analysis with machine learning approaches. *Energy* 188, 116077.
- Xing, J., Luo, K., Wang, H., Jin, T., Fan, J., 2020. Novel sensitivity study for biomass directional devolatilization by random forest models. *Energy Fuel* 34 (7), 8414–8423.
- Xu, X., Chen, R., Pan, R., Zhang, D., 2020a. Pyrolysis kinetics, thermodynamics, and volatiles of representative pine wood with thermogravimetry. *Energy Fuel* 34, 1859–1869.
- Xu, X., Chen, R., Pan, R., Zhang, D., 2020b. Pyrolysis kinetics, thermodynamics, and volatiles of representative pine wood with thermogravimetry–Fourier transform infrared analysis. *Energy Fuel* 34 (2), 1859–1869.
- Yang, X., Yuan, C., He, S., Jiang, D., Cao, B., Wang, S., 2023. Machine learning prediction of specific capacitance in biomass derived carbon materials: effects of activation and biochar characteristics. *Fuel* 331, 125718.
- Zhang, W., Zhang, J., Ding, Y., He, Q., Lu, K., Chen, H., 2021. Pyrolysis kinetics and reaction mechanism of expandable polystyrene by multiple kinetics methods. *J. Clean. Prod.* 285, 125042.
- Zhang, W., Zhang, J., Ding, Y., Zhou, R., Mao, S., 2022. The accuracy of multiple methods for estimating the reaction order of representative thermoplastic polymers waste for energy utilization. *Energy* 239, 122112.
- Zhong, Y., Ding, Y., Jiang, G., Lu, K., Li, C., 2023. Comparison of artificial neural networks and kinetic inverse modeling to predict biomass pyrolysis behavior. *J. Anal. Appl. Pyrolysis* 169, 105802.
- Zhong, H., Yu, X., Zhang, J., Jiao, L., Niu, B., Tang, R., Shi, X., Vladimirovich, V.S., Xiong, Q., 2024. Predicting spatiotemporal distributions in a bubbling fluidized bed for biomass fast pyrolysis using convolutional neural networks. *Ind. Eng. Chem. Res.* 63 (8), 3744–3754.
- Zhu, X., Li, Y., Wang, X., 2019. Machine learning prediction of biochar yield and carbon contents in biochar based on biomass characteristics and pyrolysis conditions. *Bioresour. Technol.* 288, 121527.